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PMC11085325 | Buchwald – Hartwig 胺化 | 采集状态 imported_unchecked

标题	Synthesis of 5-(Aryl)amino-1,2,3-triazole-containing 2,1,3-Benzothiadiazoles via Azide – Nitrile Cycloaddition Followed by Buchwald – Hartwig Reaction
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摘要（原文节选）

An efficient access to the novel 5-(aryl)amino-1,2,3-triazole-containing 2,1,3-benzothiadiazole derivatives has been developed. The method is based on 1,3-dipolar azide – nitrile cycloaddition followed by Buchwald – Hartwig cross-coupling to afford the corresponding N-aryl and N,N-diaryl substituted 5-amino-1,2,3-triazolyl 2,1,3-benzothiadiazoles under NHC-Pd catalysis. The one-pot diarylative Pd-catalyzed heterocyclization opens the straightforward route to triazole-linked carbazole-benzothiadiazole D-A systems. The o …

数据表 · Table 1（5 行，含数值行 5 行）

The long-wavelength absorption and emission bands, quantum yields, the frontier molecular orbitals (FMOs) energies a for compounds 4b, 5f, 6d, 6g, and 7a in DCM.

Compound	λ abs, nm	λ em, nm	QY, Air, %	HOMO, eV	LUMO, eV	Gap, eV
4b	407	562	89	− 5.54	− 3.49	2.05
5f	405	550	86	− 5.59	− 3.48	2.11
6d	405	546	95	− 5.78	− 3.48	2.30
6g	391	529	54	− 5.78	− 3.47	2.31
7a	404	540	95	− 5.83	− 3.49	2.34

脚注（逐字）：a Calculated from electrochemical data.

正文对应句（逐字）：Then, following our previous experience [31] and literature precedents [29], we performed the pre-optimization of the dipolar [3 + 2]-cycloaddition (DCR) for the reaction of BTD-nitrile 3a with benzyl azide by screening simple catalysts (K2CO3, Cs2CO3, KOtBu) and solvents (DMF, DMSO) under moderate

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